Transfer Representation Learning Meets Multimodal Fusion Classification for Remote Sensing Images

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***Abstract*— To maximize the complementary advantages of synergistic multimodal, a transfer representation learning fusion network (TRLF-Net) is proposed for multisource remote sensing images collaborative classification in this article. First, with respect to the feature encoding, we design a dual-branch attention sparse transfer module (DAST-Module), which combines the spatial and channel attention (CA) masks to migrate the advantage attributes of the panchromatic (PAN) and the MS images mutually. This not only enhances their respective image advantages but also facilitates the sparse fusion of low-level features. Second, for the separation of multiscale information, a deep dual-scale decomposition module (DDSD-Module) is designed, which allows the decompose of high-frequency and low-frequency components. Then it uses the decomposed information to make the essential difference as small as possible, and the surrounding contour difference is as large as possible of the complementary multimodal image through the design of the loss function. Finally, to address the problem of large intraclass and small interclass differences, we develop a representation fusion of the global and local features' module (RFGAL-Module). It mainly adopts global features to sort local features within classes, and then outputs them in a cascade. Thus, the characterization ability of features is improved, and the global and local features are used in a coordinated manner to accomplish the sample classification tasks. In particular, the experimental results demonstrate that TRLF-Net can obtain much improved accuracy and efficiency. The code is accessible in: https://github.com/ru-willow/SRLF-Net.**

***Index Terms*— Attention sparse transfer (AST), deep learning, dual-scale decomposition, feature representation, multiresolution classification, remote sensing.**

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I. Introduction

With the continuous advances of remote sensing imaging technology, we can easily obtain an increasing amount of spatial and spectral information from multimodal data [1]. Therefore, the fusion and analysis of multisource remote sensing images is an important and urgent research topic [2], [3]. Remote sensing sensors produce data and images with diverse characteristics, which are mainly reflected in the spatial resolution, spectral resolution, and imaging mode of the image [4]. For example, in the passive imaging mode, the panchromatic (PAN) images and RGB images have high spatial information, and the multispectral images (MSs) and hyperspectral images (HSIs) have high spectral resolution [5]. In the active imaging mode, the light detection and ranging images (LiDARs) can obtain height and intensity information [6], and the synthetic aperture radar (SAR) can obtain amplitude and phase information around the clock [7]. Zhong et al. [8] proposed a semantic allocation level multifeature fusion method, which was experimentally verified to be superior to the single-feature method.

The MS satellites can acquire both the high spatial resolution PAN images (one band) and the high spectral resolution MS images (four bands) [9]. They have important constructive significance for the description of features [10]. Furthermore, they also show great potential in the fine classification and effective identification of targets [11], [12], [13]. At present, the fusion classification of the MS and PAN images is widely used in the fields of precision agriculture, resource surveys, environmental monitoring, and national defense security [14], [15].

Regarding the fusion classification of the MS and PAN images, three frameworks have been presented, all of which is accomplished in two separate steps [16]. The first framework is the traditional components substitution (CS) fusion algorithm [17]. It replaces the decomposed components with the downsampling PAN image and then performs manual feature selection and classification [18]. The second framework is the multiresolution analysis (MRA) fusion algorithm [19]. It fuses the components obtained from multiscale decomposition and inverts them to obtain the fused image,

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followed by feature extraction and classification. With the speedy progress of deep learning, the other framework is the feature fusion and classification of deep extraction [20], [21]. It extracts the features of the dual-branch by means of nonshared parameters, and then directly fuses and classifies them [22], [23].

Recently, deep learning methods [24], [25], [26], [27] have been broadly adapted in the fusion classification of the MS and PAN images. Hong et al. [28] summarized the mainstream applications in the remote sensing field from four aspects: PAN image sharpening, hyperspectral and multispectral image fusion, multimodal feature learning, and cross-modal feature learning. Wang et al. [29] proposed a context-based global multilayer feature fusion network for deeper interpretation of remote sensing scenes. Xing et al. [30] designed a decision attention module to automatically update the combined weights of each decision to obtain the final classification category. Li et al. [31] proposed a multimodal architecture search network to solve the problem of land cover classification. Li et al. [21] proposed an adaptive multiscale deep fusion residual network, which can fully apply multiscale hierarchical features and extract features with strong discrimination. Cao et al. [32] proposed the dual-frequency attention fusion network (DFAF-Net), which can synergize frequency-aware attention and adaptive feature fusion to improve the classification accuracy. Liu et al. [33] proposed a novel cross-resolution hidden layer feature fusion network to jointly classify multiresolution MS and PAN images. However, they did not take into account the inherent characteristics of remote sensing scenes and lacked the representation of specific features.

Nowadays, our research group has also made good progress in multimodal remote sensing image fusion and interpretation. In [34], Liu et al. use deep learning methods to fuse the MS and PAN images. They propose a stacked autoencoder network (SAE) to exploit the spectral information of the MS image and use a deep convolutional neural network (DCNN) to use the spatial information of the PAN image. Then, through simple cascades, two complementary features are combined and classified. In [35], Zhu et al. first innovatively use an adaptive center offset sampling strategy (ACO-SS) to cut the input patch blocks according to the neighborhood boundaries. Then designing different attention mechanisms for different inputs to maximize images advantages. Finally, cascade fusion and classification for the multimodal input images are performed. In [36], Zhu et al. first introduce an interactive data fusion strategy (IDFS) to decrease the information distribution difference between the MS and PAN images, then they design different feature extraction modules for different dominant inputs. Finally, they gradually merge them to improve the accuracy of classification. Ma et al. [37] present a spatial–channel collaborative attention enhancement network (SCCA-Net) for multiresolution remote sensing images' classification. It combines sample and feature augmentation to improve the classification accuracy. Ma et al. [38] propose an adaptive hybrid fusion network (AHF-Net). This method can improve the fusion classification result. The effectiveness of the algorithm was verified on several datasets.

From the above analysis and discussion, we derived that the deep learning algorithms are very applicable when dealing with the MS and PAN images fusion classification problem. But there are still some problems that need to be solved. The main challenges are the following.

1) The MS and PAN images describe the common scene, but they represent two dissimilar modalities. Therefore, they have different advantage masks. If they are directly multiplied with the original features, it is not conducive to low-layer features' fusion and extraction, which decreases the performance of the network.
2) The MS and PAN images differ in resolutions, but they share the same semantic information. If the encoded features are decoded directly, it would be unfavorable to analyze the high-frequency components (contour information) and low-frequency components (semantic information).
3) Due to the sophisticated imaging principles of remote sensing sensors, the PAN and MS images often have great differences within classes and small differences between classes. This is not conducive to the representation of the global general information in the fully connected layer and the local target information in the convolutional layer of the same target.

Based on the above problem analysis, we propose a transfer representation learning fusion network (TRLF-Net) for multisource remote sensing images' classification. The major contributions are the following.

1) In terms of feature encoding, a dual-branch attention sparse transfer module (DAST-Module) that combines spatial attention (SA) and channel attention (CA) mechanisms is designed. The DAST-Module first obtains the low-level structural texture information through spectral and spatial encoding, then obtains different dominant masks through spectral attention and SA, and finally transfers the different dominant masks to enhance the representation ability of features. Therefore, this module is capable of learning the dominant masks of the input themselves through different attention mechanisms and enhancing the features of the opponent using the learned mask. Thereby, both feature enhancement and gradual fusion of the underlying features can be achieved.
2) Regarding feature decoding, considering the different resolutions between the PAN and MS images, a deep dual-scale decomposition module (DDSD-Module) is proposed before feature decoding. The DDSD-Module first obtains low-frequency energy semantics and high-frequency contour background features through dual-scale decomposition, and then characterizes the overall sharpness and contrast of the image through dual-scale feature extraction. Thence, this module realizes the decomposition of high frequency (detailed texture information) and low frequency (energy semantic information), and it can achieve feature optimization through the design of loss function, thereby improving the accuracy of classification.
3) For the representation of features, taking into account that the problems of small differences between classes

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and large differences within classes are between the PAN and MS images, a representation fusion of the global and local features' module (RFGAL-Module) is designed. The RFGAL-Module first clusters the local features according to the global features, and then applies the cosine similarity to rank local features within the class, next selects the feature space with high influence for the classification task, and finally converts it into a vector formal and global features are cascaded. This module allows the reranking of the local features based on the global features. That is, similar objects in the image are distributed in the same position to improve the feature representation ability of similar samples.

II. Related Work

In this section, we will recommend the three related works of our proposed algorithm in detail. Those are the CA and SA mechanism, multiscale decomposition, and feature representation for the multiresolution remote sensing images.

A. CA and SA Mechanism

The attention mechanism in computer vision is mainly divided into two types [39], [40]: CA mechanism [41] and SA mechanism [42]. In [36], he proposes an adaptive spatial and channel attention module (ASACA-Module). The specific operations are as follows:

$$F_{\text{out}} = \psi(f_{\text{ex}}(f_{\text{sq}}(X)) \cdot X \tag{1}$$

where $X \in \mathbb{R}^{H\times W\times C}$ is the input feature. If it is a CA mechanism, $f_{\text{sq}}(\cdot)$ is a global average pooling, and $f_{\text{ex}}(\cdot)$ is a connected layer. If it is an SA mechanism, $f_{\text{sq}}(\cdot)$ is an average pooling along the channel dimension, and $f_{\text{ex}}(\cdot)$ is a convolution layer. $\psi(\cdot)$ is an activation function (Sigmoid or Softmax). $F_{\text{out}} \in \mathbb{R}^{H\times W\times C}$ is the enhanced output feature.

The CA mechanism can find the dependencies between channels through its own learning, thereby enhancing the channels that have a greater impact on the classification task, while weakening the weight of the channels that have a less impact on the classification task. The SA mechanism can give different weights to objects with large differences in size to enhance the spatial information. But they do not consider the interaction enhancement and fusion between complementary attention masks.

B. Multiscale Decomposition

The traditional multiscale decomposition mainly uses some classic algorithms. For example, discrete wavelet transform (DWT), contour wave transform (CWT), and pyramid transform (PT). In [43], he first uses MRA to merge the decomposed components to obtain a new fused image, and then he inputs it into the convolutional neural network (CNN) for feature extraction and classifies it. Nevertheless, he adopts a non-end-to-end learning method. That is, it needs to be processed in two steps: fusion components and classification features. So it is not conducive to the adjustment of network parameters. In [44], she introduces contourlet waves into CNN, but she does not perform MRA, and only merges it with the features extracted by CNN simply and rudely. Based on the above analysis, they all do not consider adding multiscale transformation and analysis (high-frequency and low-frequency information) in deep learning to form an end-to-end network simultaneously, which not only is not conducive to the optimization and training of the entire network but also makes the network not elegant enough.

C. Feature Representation

Considering that the local features and global features have different representational meanings, we generally merge the local features and global features after feature extraction is completed. In [45], he directly merges global general information and local target information. But there are generally large differences within classes and small differences between classes on large-scale remote sensing images. For this problem, the normal operation is feature selection [46]. That is, the features with larger differences in different categories or the features with smaller differences in the same category are selected as the final classification features. But this puts forward higher requirements for feature selection and similarity measure technology.

III. Methodology

In this section, we present a novel transfer representation learning fusion framework for the PAN and MS images' classification in detail, called TRLF. It is mainly composed of DAST-Module, DDSD-Module, and the RFGAL-Module.

A. Transfer Representation Learning Fusion Network (TRLF-Net)

In the overall network architecture design (see Fig. 1), considering that the encoder–decoder structure [47] (EDS) can combine different contextual information and has a good local representation ability, we adopt it as a baseline network framework. At the same time, we insert the DAST-Module (see Section III-B), the DDSD-Module (see Section III-C), and the RFGAL-Module (see Section III-D) into it.

According to the design principles of deep neural networks, in the encoder stage, we apply the DAST-Module to transfer advantages attention attributes. The DDSD-module is to obtain the low-frequency detail information and the high-frequency background information of the two inputs after attention sparse transfer (AST), respectively. In the decoder stage, we recharacterized the local and global features by the RFGAL-Module. Next, we cascade it with the fully connected output feature. Finally, we classify it with three fully connected layers.

The encoded features undergo deep dual-scale decomposition to obtain low-frequency detail semantics feature $F_{\text{MS}}^{\text{LF}}$ and high-frequency contour background feature $F_{\text{MS}}^{\text{HF}}$ of the MS image, respectively. Similarly, $F_{\text{PAN}}^{\text{LF}}$ and $F_{\text{PAN}}^{\text{HF}}$ can also be obtained.

Regarding the output of TRLF-Net, the design of the loss function is as follows.

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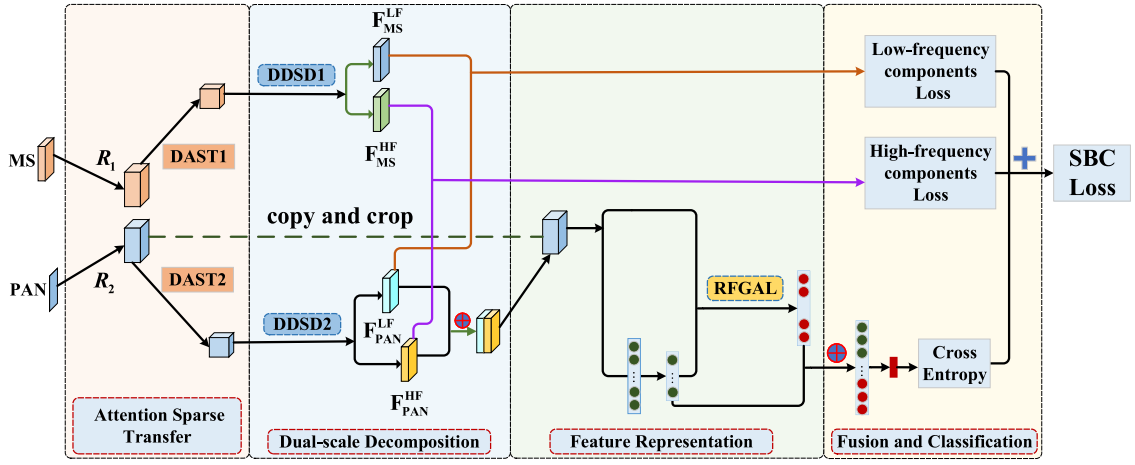

Fig. 1. TRLF-Net. ⊕ represents feature concatenation.

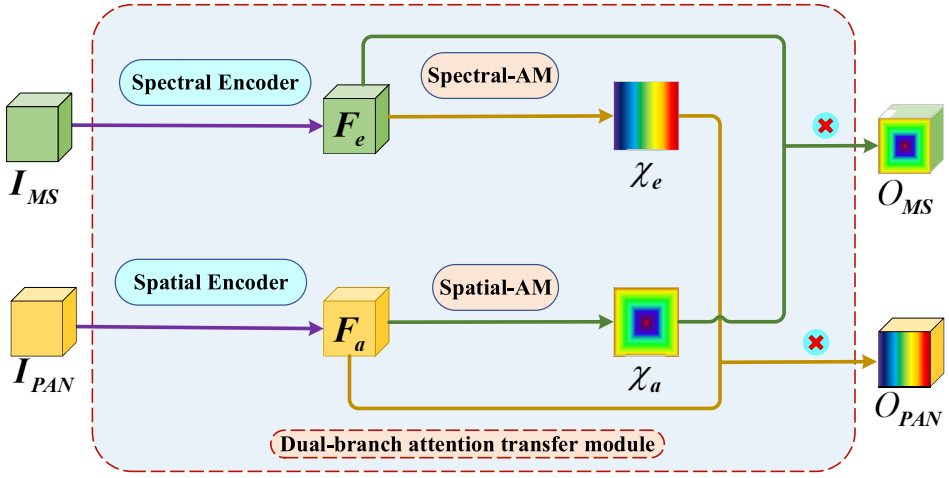

Fig. 2. DAST-Module. ⊗ represents the spatialwise multiplication operation.

1) *Low-Frequency Components' Loss*

$$L_1 = f\left(\left\|F_{\mathrm{MS}}^{\mathrm{LF}} - F_{\mathrm{PAN}}^{\mathrm{LF}}\right\|_2^2\right) \tag{2}$$

where $f(\cdot)$ is the tanh activation function, which is used to compress the feature map between (0,1).

2) *High-Frequency Components' Loss*

$$L_2 = f\left(\left\|F_{\mathrm{MS}}^{\mathrm{HF}} - F_{\mathrm{PAN}}^{\mathrm{HF}}\right\|_2^2\right). \tag{3}$$

3) *Cross-Entropy Loss*

$$\mathrm{Loss}_{\mathrm{CE}} = -\frac{1}{n_b}\sum_{i=1}^{n_b}[y_i \log(\widehat{y}_i) + (1-y_i)\log(1-\widehat{y}_i)] \tag{4}$$

where n_b represents the batch size, y_i denotes the label, while $\widehat{y}_i$ represents the category probability.

4) *Total Loss Function:* The high-frequency background contour feature map is used to derive the unique features of the feed image, and the low-frequency detailed semantic is used to extract the common features of the source image. The multispectral (MS) and the PAN images are different modalities of the same scene; in theory, their shared semantic information is the same, but their unique features are different. Therefore, we should make the difference in the background feature map greater (L_2). In contrast, the difference in detailed feature maps should be smaller (L_1). In the training process of the neural network, the goal is to reduce the loss as much as possible, thence the total loss function is designed as follows:

$$\mathrm{Loss}_{\mathrm{SBC}} = L_1 - L_2 + \mathrm{Loss}_{\mathrm{CE}}. \tag{5}$$

B. Dual-Branch Attention Sparse Transfer Module (DAST-Module)

The spectral information of the MS images is rich. Correspondingly, the spatial resolution of the PAN images is very high. That is, these two types of images have their own particularities and respective competitive advantages. Therefore, the DAST-Module we proposed is mainly to transfer their respective advantages to each other through the attention mechanism mask, thereby achieving the sparse integration of different characteristics. The specific implementation process is shown in Fig. 2.

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1) Spectral and Spatial Encoders: First, we use a dual-input way to encode the spectral and spatial features of the MS and PAN images, respectively, which can gain the low-level structural texture information

$$F_e = f_e(I_{\mathrm{MS}}) \\ F_a = f_a(I_{\mathrm{PAN}}) \tag{6}$$

where I_{MS} and I_{PAN} are the input features. $f_e(\cdot)$ and $f_a(\cdot)$ are the spectral and spatial encoders, respectively.

2) Spectral and SA Mechanisms: We perform spectral attention mechanism coding and SA mechanism coding on the features F_e and F_a to get the spectral mask χ_e and the spatial mask χ_a, respectively

$$\chi_e = f_{\mathrm{se}}(F_e) \\ \chi_a = f_{\mathrm{sa}}(F_a) \tag{7}$$

where $f_{\mathrm{se}}(\cdot)$ is spectral attention mechanism, and $f_{\mathrm{sa}}(\cdot)$ is SA mechanism.

3) Attention Transfer: To select the focus position, a more distinguishable representation feature is generated. We multiply the feature vector χ_e with the F_a to obtain a new module output feature map O_{PAN}. Similarly, we multiply the feature vector χ_a with F_e to obtain a new module output feature map O_{MS}

$$O_{\mathrm{PAN}} = \chi_e \cdot F_a \\ O_{\mathrm{MS}} = \chi_a \cdot F_e. \tag{8}$$

Through the dual-branch attention sparse transfer mechanism module (DAST-Module), the output MS feature map O_{MS} not only has the spectral information of the MS image but also integrates the spatial information of the PAN image. The output PAN feature map O_{PAN} not only has the spatial information of the PAN image but also learns the channel information of the MS image. Therefore, we transfer and integrate the superior characteristics of the other party through automatic learning, and further improve the sparse characterization ability of features, which is conducive to the classification of subsequent features [48].

This process can encode the channel information contained in the MS image as the weighting knowledge prior to each channel of the PAN image. It can calculate the significance of per feature channel by learning. Then we can refer to the weight to emphasize the valuable features and suppress features with little impact. Furthermore, the spatial information of the PAN image can also be encoded as the weighting information to each space of the MS image feature. The relative importance of each feature space can be calculated by automatic learning, and then the weight can be referred to determine the feature attention of the classification task.

C. Deep Dual-Scale Decomposition Module (DDSD-Module)

To separate the low-frequency semantic energy information from the high-frequency background contour information of the input image, we need to decompose the original features. At present, most algorithms are based on filters or optimization-based methods to decompose images. They all need to contain a certain amount of prior knowledge, and they are all manually designed decomposition algorithms. However, the image decomposition algorithm we emphasize is essentially a feature extractor. Therefore, we propose a new DDSD-Module, which uses dual-scale decomposition to extract different types of information. The specific design process is shown in Fig. 3:

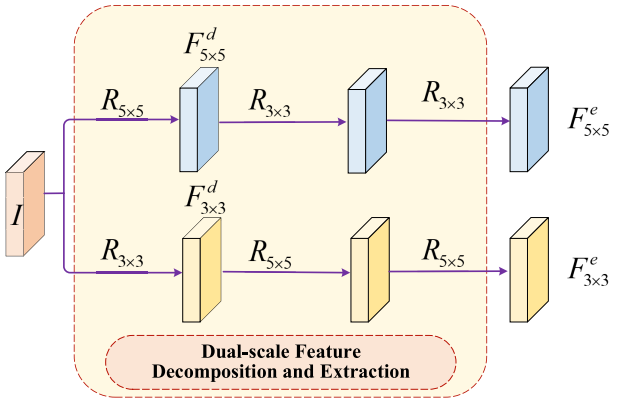

Fig. 3. DDSD-Module. $R_{i\times i}$ represents the combination of convolution operation, batch normalization (BN), and rectified linear units (ReLUs).

1) Dual-Scale Decomposition: For input feature I, we use a 5×5 convolution kernel $f_{5\times5}(\cdot)$ and a 3×3 convolution kernel $f_{3\times3}(\cdot)$ to obtain low-frequency information (detailed semantic features) $F^d_{5\times5}$ containing a large number of pixel intensity changes, and a small amount of pixel intensity changes high-frequency information (background contour features) $F^d_{3\times3}$, respectively

$$F^d_{5\times5} = f_{5\times5}(I) \\ F^d_{3\times3} = f_{3\times3}(I). \tag{9}$$

2) Dual-Scale Feature Extraction: Low frequency represents the overall average energy characteristics of the image. Therefore, for low-frequency detail features, we use a 3×3 convolution kernel to extract, which can roughly describe the overall clarity and contrast of the image. However, high frequency represents the edge change part of the image. Therefore, for the high-frequency contour feature, we use a 5×5 convolution kernel to extract, which can characterize the edge contour features in detail

$$F^e_{5\times5} = f_{3\times3}\left(F^d_{5\times5}\right) \\ F^e_{3\times3} = f_{5\times5}\left(F^d_{3\times3}\right). \tag{10}$$

Compared with the traditional MRA algorithms, this module can automatically learn the decomposed low-frequency and high-frequency information, thereby achieving the analysis and processing of extracting features. At the same time, it can not only be plug-and-play applied in the deep learning framework but also allow the evaluation of the whole network through end-to-end learning strategies.

D. Representation Fusion of the Global and Local Features Module (RFGAL-Module)

Since the intricate imaging principles of remote sensing images, the images we obtain often have the problems of

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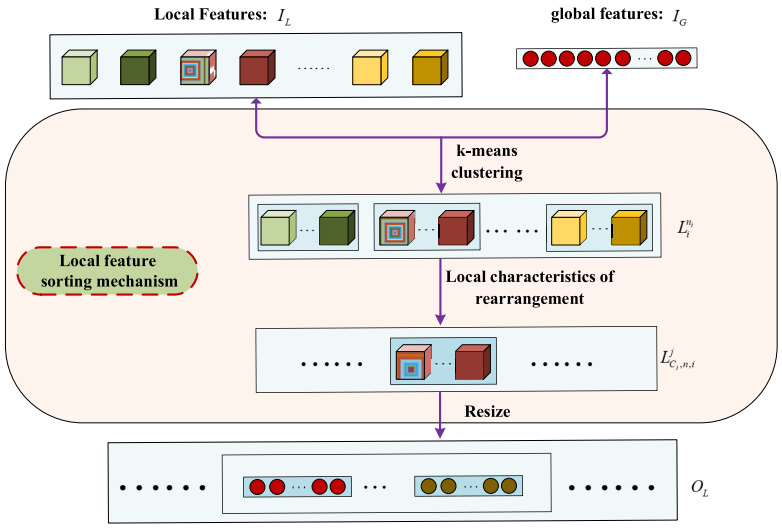

Fig. 4. RFGAL-Module.

large differences within classes and small differences between classes. This leads to the fact that when we use deep neural networks to obtain features, the global features of the two patches are similar, but the local features are different, vice versa. Therefore, we need to sort the local features and comprehensively characterize the global features and local features at the same time, which can overcome the limitation of only applying global features or local features, and then improve the classification accuracy of features. The specific implementation process is shown in Fig. 4.

As is known, the outputs of the last fully connected layer and the last convolutional layer are the global and local features of the source image, respectively. The global feature contains the general information of the image, and it is an important part of image classification. However, in addition to the global features, the local features are also significant in image classification.

1) Clustering of Local Features: First, we perform k-means [49] clustering on n global features I_G to obtain $G_1^{n_1}$, $G_2^{n_2}, \ldots, G_i^{n_i}, \ldots, G_k^{n_k}$

$$\begin{aligned} G_1^{n_1}, G_2^{n_2}, \ldots, G_i^{n_i}, \ldots, G_k^{n_k} &= g(I_G) \\ \text{s.t. } n_1 + n_2 + \cdots + n_i + \cdots + n_k &= n \end{aligned} \tag{11}$$

where $g(\cdot)$ represents the k-means clustering operation, k is the number of clusters, and $G_i^{n_i} \in \mathbb{R}^{HWC}$ means that the ith cluster has n_i global features.

Then, we find the corresponding local features $L_i^{n_i} \in \mathbb{R}^{H\times W\times C}$ $(L_{C_1,1}, \ldots, L_{C_1,n_1}, \ldots\ldots, L_{C_k,1}, \ldots, L_{C_k,n_k})$ in each pile. Next, we obtain the center sample C_1, $C_2, \ldots, C_i \in \mathbb{R}^{H\times W\times C}, \ldots, C_k$ in each pile by averaging within the pile.

2) In-Class Ordering of Local Features: Calculate the spatial cosine similarity [50] between the local features of all the samples in each heap and the central sample feature

$$\begin{aligned} \text{Sim}\langle X_i, Y_i\rangle &= \cos\langle X_i, Y_i\rangle \\ \text{s.t. } 1 \le i &\le H \times W \end{aligned} \tag{12}$$

where X_i refers to each spatial information of the local features of all the samples except the central sample in each heap, Y_i refers to each spatial information of the local features of the central sample in each heap, and $X_i = [x_1, x_2, \ldots, x_c]^T$, $Y_i = [y_1, y_2, \ldots, y_c]^T$, and c is channel of the local features.

Then, the spatial similarities are sorted from largest to smallest, and the corresponding sample space is also reorganized in the corresponding $H \times W$ space

$$L_{C_i,n,i}^j = \text{sort}(\text{Sim}\langle X_i, Y_i\rangle) \tag{13}$$

where i belongs to the position in space $H \times W$ before sorting, and j belongs to the position in space $H \times W$ after sorting. $L_{C_i,n,i}^j$ $(L_{C_1,1,i}^j, \ldots, L_{C_1,n_1,i}^j, \ldots\ldots, L_{C_k,1,i}^j, \ldots, L_{C_k,n_k,i}^j)$ represents the pixel at the ith position of the nth sample of the C_ith pile sorted to the jth position.

3) Feature Transformation: Finally, we resize the reordered local features into a vector

$$O_L = \text{resize}\left(L_{C_i,n,i}^j\right). \tag{14}$$

This module can automatically sort local features through global features, which increases the differences between classes and reduces the intraclass differences, thereby increasing the representation ability of features.

IV. Experimental Study

A. Dataset Description and Experimental Settings

In this section, four datasets are used to validate the robustness and effectiveness of the proposed method, and each of the multimodal datasets contains a pair of corresponding to the PAN and MS images.

They are the Level 1A data (Preprocessing radiometrically corrected image products). For the GaoFen-1 satellite, the PAN image has a spatial resolution of 2 m, and the MS image [R, G, B, near infrared (NIR)] has a spatial resolution of 8 m. For the GaoFen-2 satellite, the PAN image has a spatial

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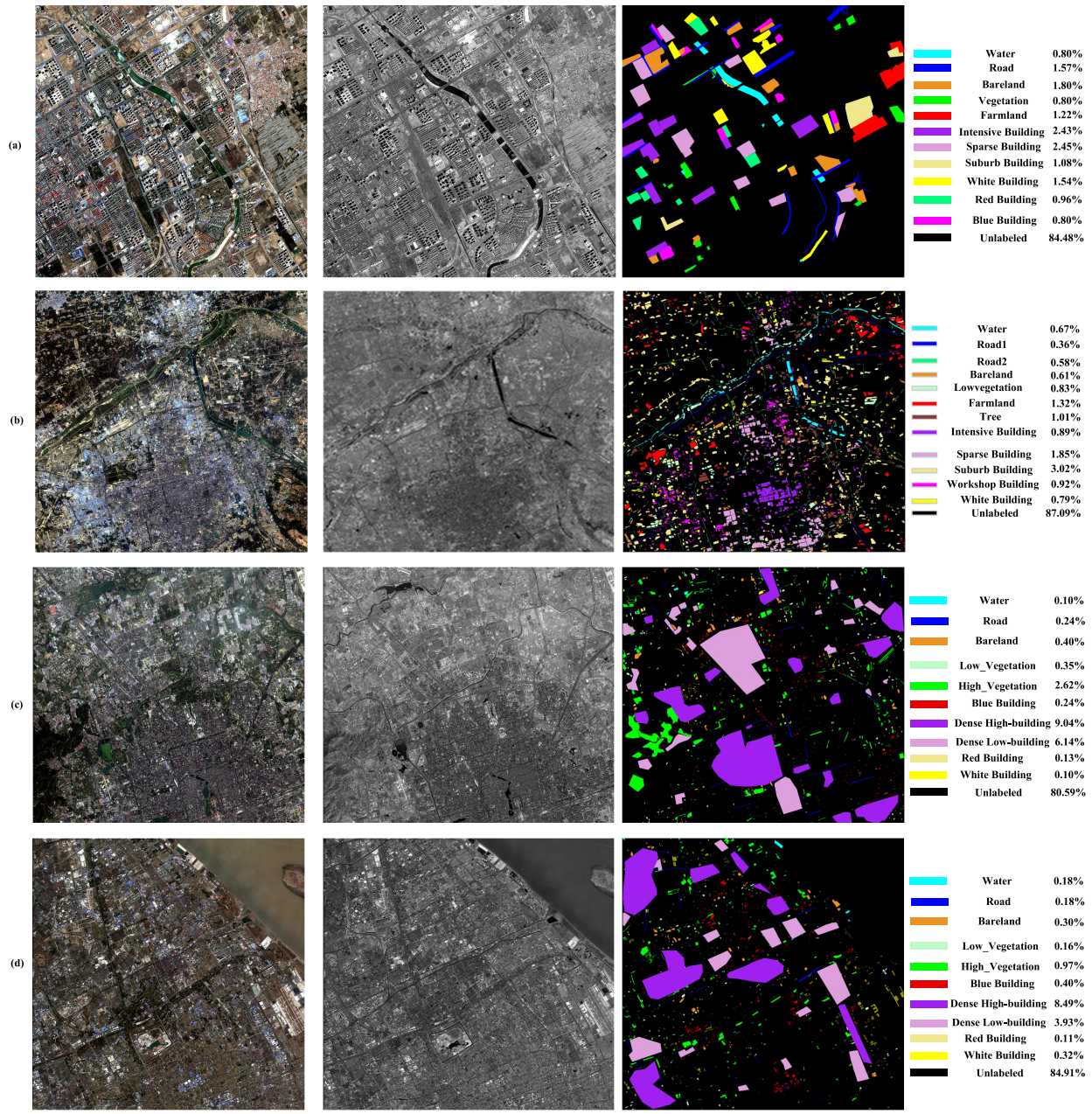

Fig. 5. Dataset. In the first column is the MS images. In the second column is the PAN images. In the third column are the ground-truth images. In the last column are the category labels. (a) Hohhot images. (b) Xi'an images. (c) Beijing images. (d) Shanghai images.

resolution of 1 m, and the MS image has a spatial resolution of 4 m. They are bought by our laboratory.

1) Hohhot images [see Fig. 5(a)] were acquired over Hohhot, China (111°39′30″W, 40°49′00″N). The PAN image contains 8004 × 8404 pixels, and the MS image is 2001 × 2101 pixel-sized. They contain 11 types of scenes: water, road, bareland, vegetation, farmland, and six kinds of building areas (intensive building, sparse building, suburb building, white building, red building, and blue building).
2) Xi'an images [see Fig. 5(b)] were acquired on August 29, 2015, over Xi'an, China (108°56′52″W, 34°15′14″N). The PAN image contains 18 192 × 18 164 pixels, and the MS image is 4548 × 4544 pixel-sized. They contain 12 types of scenes: water, two kinds of roads, bareland, low vegetation, farmland, tree, and five kinds of building areas (intensive building, sparse building, suburb building, workshop building, and white building).
3) Beijing images [see Fig. 5(c)] were acquired over Beijing, China. The PAN image contains 29 200 × 27 620 pixels, and the MS image is 7300 × 6905 pixel-sized. They contain ten types of scenes: water, road, bareland, low vegetation, high vegetation, and five types of building areas (blue building, dense high building, dense low building, red building, and white building).
4) Shanghai images [see Fig. 5(d)] were acquired over Shanghai, China. The PAN image contains 29 200 × 27 620 pixels, and the MS image is 7300 × 6905 pixel-sized. They contain ten types of scenes: water, road,

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TABLE I
HYPERPARAMETERS' DESIGN OF NETWORK

Types	Input	PAN_Branch	MS_Branch	Output
prejust	$4 \times 16 \times 16$ $1 \times 64 \times 64$	$\begin{bmatrix} conv2d: 3\times 3 \\ BatchNorm \\ Relu \end{bmatrix}$,$1 \to 8$	$upsample: 16 \to 64$ $\begin{bmatrix} conv2d: 3\times 3 \\ BatchNorm \\ Relu \end{bmatrix}$,$4 \to 8$	$8 \times 64 \times 64$ $8 \times 64 \times 64$
Encoder AST	$8 \times 64 \times 64$ $8 \times 64 \times 64$	$\begin{bmatrix} \begin{bmatrix} conv2d: 3\times 3 \\ BatchNorm \\ Relu \end{bmatrix} \times 2 \\ \text{DAST} \\ \text{maxpool:} 2\times 2 \end{bmatrix} \times 4$	$\begin{bmatrix} \begin{bmatrix} conv2d: 3\times 3 \\ BatchNorm \\ Relu \end{bmatrix} \times 2 \\ \text{DAST:} \\ maxpool: 2\times 2 \end{bmatrix} \times 4$	$128 \times 4 \times 4$ $128 \times 4 \times 4$
Dual-scale Decomposition	$128 \times 4 \times 4$ $128 \times 4 \times 4$	DDSD1 $128 \times 4 \times 4 \to 128 \times 4 \times 4$	DDSD2 $128 \times 4 \times 4 \to [64 \times 4 \times 4] \times 2$	$128 \times 4 \times 4$ $[64 \times 4 \times 4] \times 2$
Decoder	$128 \times 4 \times 4$	$\begin{bmatrix} convtransposed2d \\ concatenating \\ \begin{bmatrix} conv2d: 3\times 3 \\ BatchNorm \\ Relu \end{bmatrix} \times 2 \end{bmatrix} \times 4$	\	$8 \times 64 \times 64$
RFGAL Fusion and Classification	$8 \times 64 \times 64$	$conv2d: 3\times 3, 8 \to 3$ $fc1: [12288, 1024]$ $fc2: [1024, 512]$ RFGAL:12288 + 512 = 12800 $fc3: [12800, classes]$	\	classes

bareland, two kinds of vegetation, and five kinds of building areas (blue building, two kinds of dense building, red building, and white building).

The proposed TRLF-Net is trained with PyTorch of the deep learning framework. All the experiments are running on AMAX Information Technologies Inc. (AMAX) Workstation with GTX1080Ti X 8 GB graphics processing unit (GPU) and 128-GB RAM under Ubuntu 16.04 long term support (LTS) in this article.

Considering the PAN and MS image's resolution and the size of the actual terrain, the input sizes are $64 \times 64 \times 1$ and $16 \times 16 \times 4$ pixels. About 2% and not less than 1500 labeled data are selected as training samples from each category. The intersection-over-union (IOU) ratio [51] of the training and test samples is less than 1/4.

The initial learning rate is 0.002, the batch size is 64, and the epoch number is 40. Furthermore, the network architecture parameter devise is shown in Table I.

B. Selection of Hyperparameters: K

K-means is a relatively simple clustering algorithm because it only needs to choose a suitable K value to represent the number of clusters. But for classification tasks, the selection of the K value is very important. For different values of K ($1 < K \leq$ classes), the evaluation indicators are the overall classification accuracy (OA) and kappa coefficient statistic (Kappa) on all the datasets.

Regardless of the dataset, the results (OA and Kappa) of $K = 8$ are the highest (see Figs. 6 and 7). This is mainly due to the fact that the K value is too large or too small, which will not characterize the distribution characteristics of the data. That is, the local features are not sorted, and the problem of large intraclass variation and small interclass variation cannot be solved. Therefore, in the following experiments in this article, the value of K is 8.

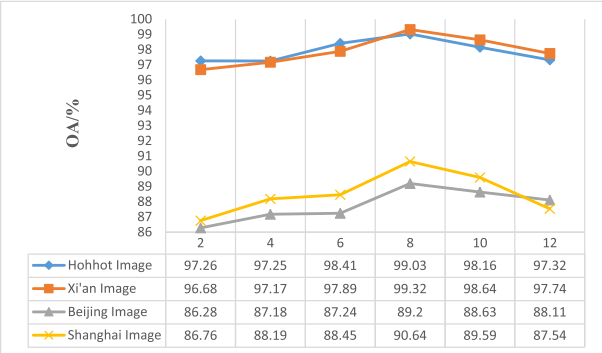

Fig. 6. Evaluation of hyperparameters: OA. K is 2, 4, 6, 8, 10, and 12, respectively.

C. Module Comparison Experiments

For the module comparison, we mainly compare the classification precision of the [36] ASACA-Module with the proposed DAST-Module, MRA module [43] (MRA-Module) with DDSD-Module, and the direct fusion of the global and local features [45] (DFGAL) with the representation fusion

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TABLE II
MODULE COMPARISON RESULTS ON THE HUHHOT IMAGE

Modules	ASACA [36]	DAST	MRA-Module [43]	DDSD-Module	DFGAL-Module [45]	RFGAL-Module
c_1: water	99.42	99.74	99.74	99.81	99.83	99.89
c_2: road	91.84	94.36	97.65	92.37	95.21	96.71
c_3: bareland	94.49	96.91	96.81	96.21	95.79	98.31
c_4: vegetation	88.27	98.07	93.99	95.61	97.11	98.59
c_5: farmland	95.39	95.78	95.12	96.21	93.63	97.40
c_6:Intensive Building	93.89	98.08	96.98	97.79	95.61	98.37
c_7: Sparse Building	97.74	97.27	96.05	99.17	98.46	97.35
c_8: Suburb Building	97.12	98.90	92.17	98.22	97.62	98.17
c_9: White Building	96.09	97.64	96.38	97.68	97.16	98.81
c_{10}: Red Building	96.17	95.03	93.40	96.01	96.62	98.62
c_{11}: Blue Building	92.80	95.73	97.25	98.67	90.74	95.05
OA(%)	94.74	96.93	95.96	96.81	95.85	97.92
AA(%)	94.46	96.67	95.78	96.40	95.56	97.71
Kappa(%)	94.11	96.56	95.48	96.42	95.35	97.67
time(s)	266.65	194.39	288.86	285.68	258.93	204.67

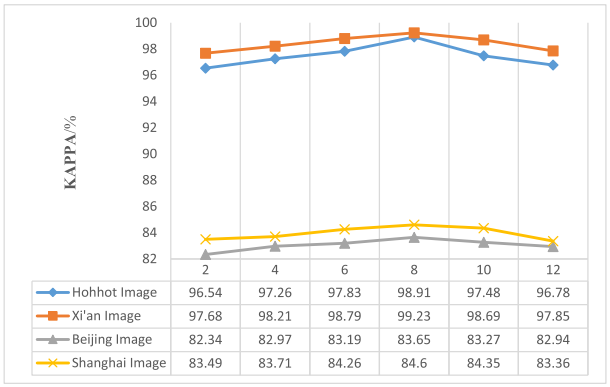

Fig. 7. Evaluation of hyperparameters: Kappa. K is 2, 4, 6, 8, 10, and 12, respectively.

of the global and local features (RFGAL) on the randomly selected Huhhot image. The evaluation indicators are OA, average accuracy (AA), Kappa, and time (s).

1) DAST-Module Versus ASACA-Module: The DAST-Module is superior to the ASACA-Module (see Table II). This is primarily explained by the fact that the ASACA-Module can only simply realize the enhancement of spatial and spectral information separately, and then integrate the enhanced features. This is equivalent to letting the feature extraction and classification network study a strong mapping relationship, and it is not conducive to the classification of samples. However, DAST-Module can realize the migration and fusion of dominant attributes while enhancing the features. That is, it can enhance its own features through masks learned from complementary sources. This module can not only realize the migration of dominant masks and the fusion of features but also make the fused features have spatial and spectral information with diverse advantages, which is conducive to the characterization of features, thus upgrading the accuracy of classification.

2) DDSD-Module Versus MRA-Module: From Table II, the precision of the DDSD-Module is higher than that of the MRA-Module. Since the DDSD-Module can realize multiresolution decomposition and analysis end-to-end, this is conducive to the adjustment of network parameters and reduces the complexity of the model. While the MRA-Module needs to first perform multiresolution decomposition on the original image, and then use certain rules to fuse high-frequency and low-frequency components. Next, we perform an inverse transformation to generate a fused image, and finally input it into a feature extraction and fusion network for sample analysis and classification. The quality of the fused image directly affects the accuracy of the classification, so the MRA-Module places higher demands on the quality of the generated fusion images.

3) RFGAL-Module Versus DFGAL-Module: On the remote sensing dataset, the results of the RFGAL-Module are higher than that of the DFGAL-Module. Because the RFGAL-Module takes into account the problems of large differences within classes and small differences between classes on remote sensing images. As we all know, the classification of deep learning mainly uses the similarity between samples feature. If we directly use the DFGAL-Module to fuse the local and global features, it will not be conducive to the characterization of intraclass and interclass sample features. However, the RFGAL-Module can use global features to sort local features, thereby reducing the difference in samples within a class, and increasing the difference in samples between classes, which is conducive to the classification of remote sensing images.

D. Ablation Study: The Establishment of Network Architecture

To demonstrate the validity of the proposal TRLF-Net, we also use the Hohhot image to prove the contribution of each part of the network. Table III shows that the results of TRLF-Net composed of DAST-Module ∪ DDSD-Module ∪ RFGAL-Module ∪ EDS are the highest.

First, the DAST-Module can use the advantage mask learned by complementary sources to enhance its own features, so that the enhanced features have diverse information.

Second, the DDSD-Module can be migrated to the deep network framework to realize the separation of low-frequency semantic and high-frequency contour information. The loss function is then designed to make the resemblance of the complementary input kernel semantic information as large

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TABLE III
CLASSIFICATION RESULTS OF DIFFERENT COMBINATIONS OF ARCHITECTURES ON THE HUHHOT IMAGE

Input-Image	DAST-Module	DDSD-Module	RFGAL-Module	EDS	OA(%)	AA(%)	Kappa(%)
√				√	97.14	97.15	97.68
√	√			√	98.04	97.41	97.62
√	√	√		√	98.84	98.23	98.36
√	√	√	√	√	99.03	98.93	98.91

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT COMBINATIONS OF LOSS ON THE HUHHOT IMAGE

High-Frequency Loss	Loss-Frequency Loss	Cross-Entropy Loss	OA(%)	AA(%)	Kappa(%)
√			96.25	96.49	96.37
	√		96.48	96.85	96.74
		√	97.56	96.17	97.64
√	√		98.75	97.49	98.34
	√	√	98.34	98.16	98.29
√	√	√	99.03	98.93	98.91

as possible, and the difference in the contour background information is as large as possible to optimize the feature extraction.

Next, the RFGAL-Module sorts the local features so that the global and local features of the same class samples are similar, while the features of samples between classes are quite different. Finally, they are inserted into EDS to form a dual-branch TRLF-Net, which is conducive to the recharacterization and fusion of features and improves the classification accuracy of remote sensing images.

Finally, Table IV shows that the result of high-frequency loss ∪ low-frequency loss ∪ cross-entropy loss is the highest, because the high-frequency loss aims to extract more refined edge features, and the low-frequency loss focuses on extracting the global average feature. Both the parts aim to improve the representational power of features through the constraints of unsupervised loss. The cross-entropy loss is to constrain the gap between the prediction and the ground truth in a supervised way, thereby improving the accuracy of classification.

E. Experiment Results and Comparison Algorithms

The proffered TRLF-Net approach is compared with five state-of-the-art methods, namely, deep multiple instance learning (DMIL) [34], ACO-SS + dual-branch fusion attention (DBFA)-Net [35], spatial-channel progressive fusion (SCPF)-ResNet [36], spatial-channel progressive fusion (SCCA)-Net [37], and AHF-Net [38]. All the parameters are followed as set in the original article to get the most optimal presentation of the outcomes.

1) Experiment Results and Analysis on Hohhot Area: For Hohhot Area, Table V and Fig. 8 show the quantitative and qualitative results.

DMIL [34] uses SAE to obtain the spectral characteristics of the MS image and uses DCNN to extract the spatial characteristics of the PAN image. Then they are flattened and cascaded, respectively. Finally, they are inputted into the fully connected layer for classification. This cascading method is simple and rude, which is not conducive to the extraction of fusion features. Therefore, DMIL is inferior to TRLF-Net in terms of accuracy and visual effects of ground truth.

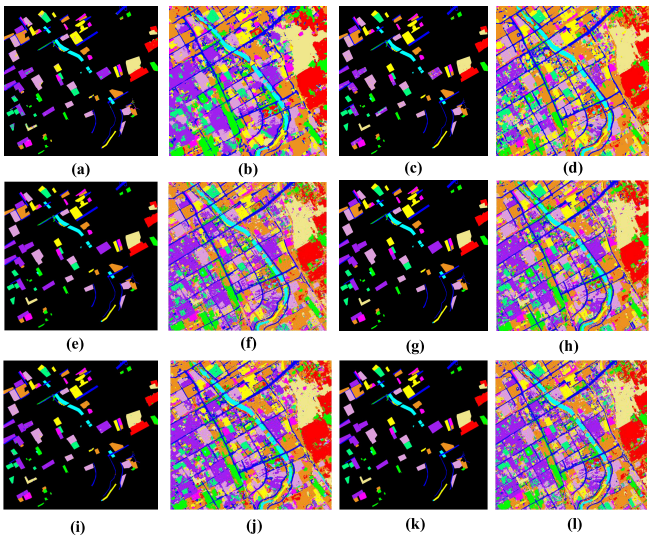

Fig. 8. Map of classification results of different methods on the Hohhot image. Each method has both ground truth and overall image results. (a) and (b) DMIL [34]. (c) and (d) ACO-SS + DBFA-Net [35]. (e) and (f) SCPF-ResNet [36]. (g) and (h) SCCA-Net [37]. (i) and (j) AHF-Net [38]. (k) and (l) TRLF-Net.

ACO-SS + DBFA-Net [35] proposes a dual-branch fusion deep network (DBFA-Net) for multiresolution image classification. Compared with DMIL, the results of ACO-SS + DBFA-Net have achieved a certain improvement, which is mainly reflected in c_4: vegetation (92.35% → 97.60%), c_5: farmland (91.91% → 99.97%), and c_{11}: blue building (92.38% → 97.51%). However, the results of some categories are not good, such as c_6: intensive building (91.14% versus 92.40%).

SCPF-ResNet [36] designs different feature extraction modules for different input images. For the MS image, a CA mechanism is designed, which is conducive to the information exchange between channel streams. For the PAN image, an SA mechanism is designed, which can ensure that objects of different sizes have different weights in different

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TABLE V
QUANTITATIVE RESULTS ON THE HOHHOT IMAGE

Algorithms	DMIL [34]	ACO-SS+DBFA-Net [35]	SCPF-ResNet [36]	SCCA-Net [37]	AHF-Net [38]	TRLF-Net
c_1: water	95.61	99.21	99.69	98.84	**99.22**	99.12
c_2: road	91.25	92.24	97.21	94.72	94.35	**98.10**
c_3: bareland	92.53	94.04	97.30	95.37	96.25	**98.83**
c_4: vegetation	92.35	97.60	98.48	96.24	96.81	**99.26**
c_5: farmland	91.91	**99.97**	97.53	98.09	98.52	99.76
c_6: Intensive Building	92.40	91.14	97.64	96.12	97.23	**99.40**
c_7: Sparse Building	97.21	96.11	98.54	98.84	98.10	**99.19**
c_8: Suburb Building	98.30	96.42	98.65	96.57	97.83	**99.39**
c_9: White Building	92.15	92.60	97.96	97.63	96.22	**98.98**
c_{10}: Red Building	84.09	92.27	97.61	96.93	96.15	**98.96**
c_{11}: Blue Building	92.38	97.51	97.63	98.11	95.69	**98.55**
OA(%)	92.60	94.80	97.86	96.80	96.82	**99.03**
AA(%)	92.74	95.38	97.78	96.95	96.64	**98.93**
Kappa(%)	91.72	94.18	97.61	96.42	96.27	**98.91**
time(s)	322.94	454.21	222.23	386.28	412.18	**185.05**

spaces. It is beneficial to the feature extraction of remote sensing images. At the same time, SCPF-ResNet adopts the idea of gradual fusion, which is conducive to the classification of features. Compared with ACO-SS + DBFA-Net, it still has made some improvements, mainly in c_{10}: red building (92.27% → 97.61%).

SCCA-Net [37] combines sample enhancement and feature enhancement to increase the classification precision. For sample enhancement, an adaptive neighborhood transfer sampling strategy is proposed, which enables each patch to adaptively transfer the neighborhood range. For feature enhancement, a local spatial and global CA module is designed, which can highlight the spatial information of the PAN image and the multichannel advantage of the MS image. Simultaneously, the interaction between the two modules not only reduces the differences between feature representations but also facilitates the fusion and classification of features. Therefore, it performs well in certain categories, such as c_5: farmland (98.09%) and c_{11}: blue building (98.11%).

AHF-Net [38] uses an adaptive weighting intensity-hue-saturation transform (IHS) algorithm, and it decreases the variance between the PAN and MS images through information sharing. At the same time, it also uses a correlation-based attention feature fusion module, which can upgrade the recognition rate of the fused features according to the importance of the features, thereby improving the classification accuracy. This is mainly reflected in c_1: water (99.22%).

The proposed TRLF-Net takes into account that the input images have different advantageous attributes, and it designs a DAST-Module to adaptively enhance its own characteristics through the opponent's advantageous mask. For MRA, it designs a DDSD-Module to optimize feature extraction. Finally, considering that the PAN and MS images have large differences within classes and small differences between classes, the proposed RFGAL can sort local features, thereby improving the global and local features' recharacterization capabilities.

Compared with SCPF-ResNet, SCCA-Net, and AHF-Net, TRLF-Net has achieved relatively high results in many categories. But in the c_5 category (farmland), the accuracy of ACO-SS + DBFA-Net is upper than that of TRLF-Net (99.97% versus 99.76%). This is mainly because the farmland is intricate, the characteristics of different farmland in different seasons are very different and the edge segmentation is fuzzy. But ACO-SS + DBFA-Net can adaptively separate samples according to the neighborhood information, so it has achieved good accuracy in c_5.

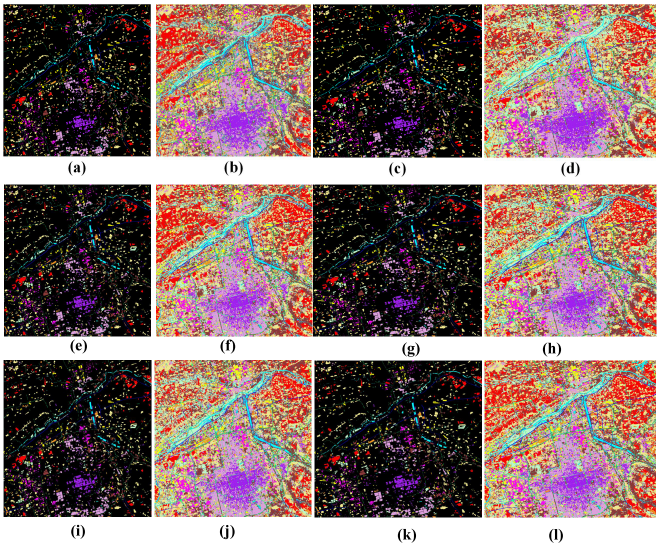

Fig. 9. Map of classification results of different methods on the Xi'an Image. (a) and (b) DMIL [34]. (c) and (d) ACO-SS + DBFA-Net [35]. (e) and (f) SCPF-ResNet [36]. (g) and (h) SCCA-Net [37]. (i) and (j) AHF-Net [38]. (k) and (l) TRLF-Net.

The run time of TRLF-Net is the shortest (see Table V). It is mainly owing to the simple structure of TRLF-Net, the number of layers is relatively shallow, and each stage feature maps are little. However, ACO-SS + DBFA-Net has more layers and feature maps, and the feature size before classification is larger, so it has the longest test time.

2) Experiment Results and Analysis on Xi'an Area: Table VI and Fig. 9 show the quantitative and qualitative results on Xi'an Area.

The c_3 (road2), c_4 (bareland), c_5 (low vegetation), and c_8 (intensive building) are very hardly distinguishable in all the classes as found through observing the confusion matrix.

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TABLE VI
QUANTITATIVE RESULTS ON THE XI'AN IMAGE

Algorithms	DMIL [34]	ACO-SS+DBFA-Net [35]	SCPF-ResNet [36]	SCCA-Net [34]	AHF-Net [38]	TRLF-Net
c_1: water	97.94	98.78	99.57	98.75	99.02	**99.94**
c_2: road1	83.86	97.96	98.41	97.01	98.54	**99.36**
c_3: road2	85.48	92.92	97.08	96.33	93.12	**98.98**
c_4: bareland	76.93	92.04	94.92	97.78	96.07	**98.99**
c_5: lowvegetation	76.10	85.31	93.93	89.56	94.49	**98.03**
c_6: farmland	90.49	98.94	99.08	99.01	99.09	**99.64**
c_7: tree	88.83	98.85	98.91	96.30	95.93	**99.26**
c_8: Intensive Building	91.00	93.99	94.52	**99.79**	97.41	98.92
c_9: Sparse Building	96.33	97.61	96.85	96.55	97.89	**98.98**
c_{10}: Suburb Building	97.83	95.76	99.18	**99.80**	97.96	99.53
c_{11}: Workshop Building	96.82	96.84	99.04	96.79	98.14	**99.53**
c_{12}: White Building	92.35	98.12	95.85	96.74	96.38	**99.53**
OA(%)	91.91	95.91	97.55	97.13	96.59	**99.32**
AA(%)	89.50	95.59	97.27	96.53	96.18	**99.22**
Kappa(%)	90.82	94.70	97.23	96.74	95.91	**99.23**
time(s)	900.56	2125.78	507.55	1061.31	1252.16	**412.18**

TABLE VII
QUANTITATIVE RESULTS ON THE BEIJING IMAGE

Algorithms	DMIL [35]	ACO-SS+DBFA-Net [36]	SCPF-ResNet [37]	SCCA-Net [38]	AHF-Net [39]	TRLF-Net
c_1: water	78.05	78.59	84.07	85.94	88.10	**90.83**
c_2: road	**89.07**	70.44	80.79	55.80	80.17	84.27
c_3: bareland	78.45	70.16	87.20	77.39	82.17	**89.81**
c_4: lowvegetation	72.81	82.13	78.05	**95.91**	87.43	82.82
c_5: highvegetation	61.60	**76.63**	59.13	61.40	66.02	70.09
c_6: Blue Building	88.31	85.10	90.21	90.81	91.13	**91.65**
c_7: Dense High-building	81.71	88.97	84.98	91.90	85.29	**90.94**
c_8: Dense Low-building	87.92	78.03	89.40	77.14	85.23	**91.11**
c_9: Red Building	84.19	72.62	69.34	54.53	80.63	**84.52**
c_{10}: White Building	79.47	79.93	53.14	75.48	79.67	**82.09**
OA(%)	82.87	84.21	86.24	84.29	85.95	**89.20**
AA(%)	70.84	77.46	77.63	76.63	76.54	**80.44**
Kappa(%)	75.41	76.32	79.08	76.86	78.65	**83.65**
time(s)	4452.36	4983.20	4328.16	4486.23	3002.40	**2897.49**

Because c_3 (road2) in downtown Xi'an is bumpy and complex. Then, c_4 (bareland) and c_5 (low vegetation) are similar in autumn and winter. Finally, the roof of c_8 (intensive building) will also change with the sun. However, our method can perform multiscale analysis. At the same time, it can integrate the characterization ability of features by sorting local features using global features, which is conducive to the essential distinction of similar categories. Therefore, whether it is a variety of roads or similar surface features caused by environmental changes, etc. None of these can affect the robustness and stability of TRLF-Net. The TRLF-Net produces remarkably high output in most categories, and this is mainly because our method can improve the feature's ability to characterize categories from both the data itself and feature extraction, and further improve the accuracy of classification. But in category c_8: intensive building and c_{10}: suburb building, the result of SCCA-Net is higher than that of TRLF-Net (99.79% versus 98.92%) and (99.80% versus 99.53%). This is mainly because the contour and color information of the building is similar, but SCCA-Net can specifically select suitable training sample blocks for adjacent similar features.

3) Experiment Results and Analysis on Beijing Area: For Beijing Area, Table VII and Fig. 10 show the quantitative and qualitative results.

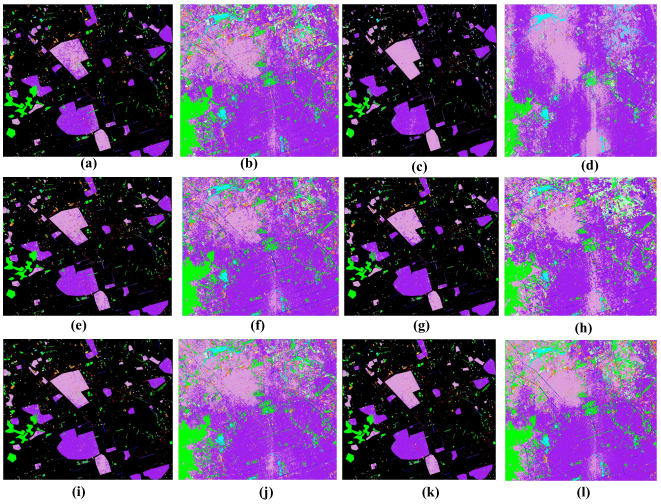

Fig. 10. Map of classification results of different methods on the Beijing image. (a) and (b) DMIL [34]. (c) and (d) ACO-SS + DBFA-Net [35]. (e) and (f) SCPF-ResNet [36]. (g) and (h) SCCA-Net [37]. (i) and (j) AHF-Net [38]. (k) and (l) TRLF-Net.

The overall results of TRLF-Net are outperformed by other methods, which is mainly due to the fact that our algorithm has superior accuracy than DMIL, ACO-SS + DBFA-Net, SCPF-ResNet, SCCA-Net, and AHF-Net in c_1: water

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TABLE VIII
QUANTITATIVE RESULTS ON THE SHANGHAI IMAGE

Algorithms	DMIL [35]	ACO-SS+DBFA-Net [36]	SCPF-ResNet [37]	SCCA-Net [38]	AHF-Net [39]	TRLF-Net
c_1: water	61.35	88.40	85.41	83.49	**91.24**	89.26
c_2: road	91.65	91.47	87.75	78.79	83.83	**91.99**
c_3: bareland	90.08	**90.85**	84.26	89.55	84.83	88.83
c_4: lowvegetation	86.50	86.63	78.25	**93.62**	82.82	81.90
c_5: highvegetation	84.73	84.91	84.23	71.74	84.95	**88.93**
c_6: Blue Building	64.46	84.48	82.38	77.96	82.37	**86.07**
c_7: Dense High-building	82.93	85.19	88.35	85.44	85.67	**92.50**
c_8: Dense Low-building	77.28	76.95	80.82	84.88	79.41	**87.91**
c_9: Red Building	82.23	92.70	80.86	92.35	81.43	**93.21**
c_{10}:White Building	93.79	93.38	94.21	84.26	93.87	**96.78**
OA(%)	80.09	83.48	85.78	84.40	84.12	**90.64**
AA(%)	76.11	75.48	85.03	84.51	83.41	**88.95**
Kappa(%)	67.10	72.32	76.61	73.69	73.59	**84.60**
time(s)	4697.48	3918.69	4925.83	4937.96	3893.90	**1187**

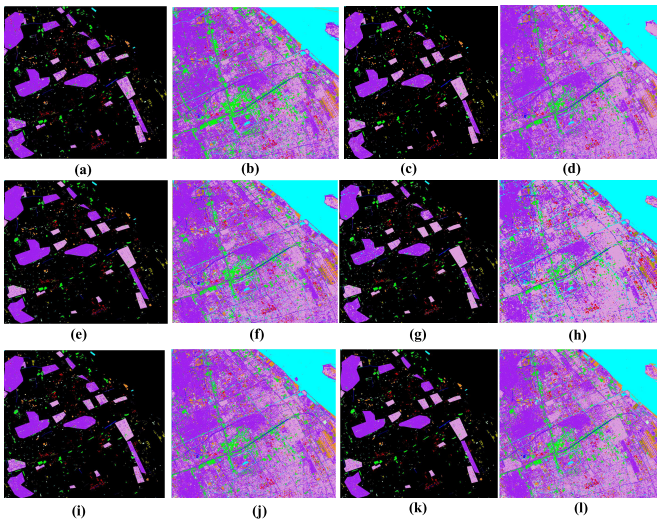

Fig. 11. Map of classification results of different methods on the Shanghai image. (a) and (b) DMIL [34]. (c) and (d) ACO-SS + DBFA-Net [35]. (e) and (f) SCPF-ResNet [36]. (g) and (h) SCCA-Net [37]. (i) and (j) AHF-Net [38]. (k) and (l) TRLF-Net.

(90.83% versus 78.05% versus 78.59% versus 84.07% versus 85.94% versus 88.10%), in c_3: bareland (89.81% versus 78.45% versus 70.16% versus 87.20% versus 77.39% versus 82.17%), and all buildings ($c_6 - c_{10}$). This is mainly because our proposed TRLF-Net aims to improve the characterization ability of features, whether it is from the encoding and decoding of features, or the fusion recharacterization of the final features. Therefore, even for categories with subtle differences, the proposed TRLF-Net also can correctly classify them. But in c_2: road, our method TRLF-Net is not as good as DMIL; in c_4: low vegetation and in c_5: high vegetation, the results of SCCA-Net and ACO-SS + DBFA-Net are also higher than TRLF-Net. This is mainly because DMIL can automatically learn the coding form of the feature. While ACO-SS + DBFA-Net and SCCA-Net can combine the information of the neighborhood to select the training sample pair, thereby achieving excellent results in the category of similar color information.

4) Experiment Results and Analysis on Shanghai Area: For Shanghai Area, Table VIII and Fig. 11 show the quantitative and qualitative results.

There is relatively more water and all the buildings are marked in detail on the Shanghai Image set. So c_1: water and $c_6 - c_{10}$: buildings belong to simple categories. Since the difference between each category is large, our proposed method TRLF-Net has achieved higher results overall. But our result is lower than AHF-Net in c_1: water (89.26% versus 91.24%), ACO-SS + DBFA-Net in c_3: bareland (88.83% versus 90.85%), and SCCA-Net in c_4: low vegetation (81.90% versus 93.62%). This is mainly because AHF-Net can perform IHS decomposition on the original image. Compared with RGB, this is more in line with computer vision recognition. ACO-SS + DBFA-Net can combine the spatial spectrum characteristics to classify multiseason bareland. SCCA-Net can adaptively cut adjacent categories into blocks and has reduced the redundancy between adjacent categories when building a training sample database.

From the comparison experiments of the above four datasets, we can find that the experimental results of the latter two datasets are significantly lower than the first two. This is mainly due to the larger size of the latter two datasets and the mislabeling of category labels in ground truth.

V. CONCLUSION

In this article, a TRLF-Net for multisource remote sensing image collaborative classification is presented. The DAST-Module combines the spectral attention mechanism with the SA mechanism. It can transfer the masks of the PAN and MS images to each other, which can achieve the fusion of different advantages and characteristics, thereby improving their respective image quality. The DDSD-Module enables to separate the low-frequency and high-frequency components of the input image. It optimizes the extraction of input data features of different modalities in the same scene through the design of the loss function. Finally, the RFGAL-Module solves the issue of large intraclass variance and small interclass variance in complex remote sensing images using global features to rank local features. Through experimental verification, the proposed TRLF-Net has achieved satisfactory results whether it is a qualitative expression or a quantitative description. In the near future, we will propose a fusion classification network based on the present work, which will be suitable for the PAN and MS images and other complementary multimodal data.

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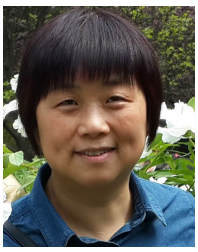

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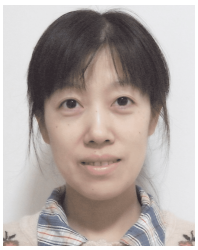

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